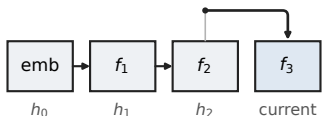
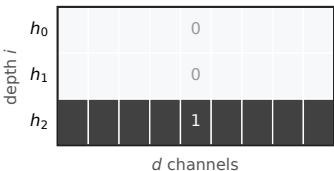


Standard pre-norm



no routing — history buried in the sum

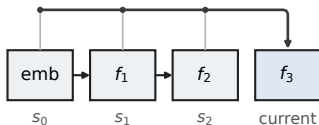


d channels

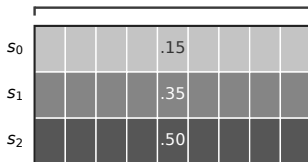
$$h_\ell = h_{\ell-1} + f(\text{LN}(h_{\ell-1}))$$

reads only the previous state

Attention Residuals (Kimi '25)



one shared $q \in \mathbb{R}^d$

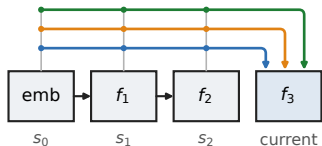


d channels

$$\alpha_i = \text{softmax}_i(q^\top \tilde{s}_i)$$

one α for all d channels (rank-1)

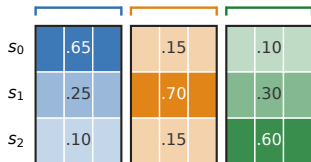
MHAR (ours)



q_1

q_2

q_3



$H = 3$ heads, d/H channels each

$$\alpha_i^{(h)} = \text{softmax}_i(q_h^\top \tilde{s}_{i, [h]})$$

independent α per d/H subspace

$W[i, c]$: read weight of depth row i into channel c of the current input (darker = larger); $\tilde{s}_i = \text{RMSNorm}(s_i)$

MHAR = same query reshaped $(d,) \rightarrow (H, d/H)$ — 0 new params, same FLOPs; $H = 1$ recovers Attn. Residuals